

CO5_ASSESSMENT 1
CONSTRAINT -BASED PROBLEM SOLVING

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1. Real-Time Face Recognition for Campus Security

A university wants to deploy a camera-based security system at the entrance that can identify registered students and staff in real time.

The system should:

- Capture faces from live CCTV/video streams.
- Detect faces under different lighting conditions.
- Extract suitable facial features.
- Recognize registered individuals.
- Maintain real-time performance on a low-cost computer.
- Minimize false recognition.

Design a complete OpenCV-based face recognition pipeline from image acquisition to final identification. Explain the preprocessing, face detection, feature extraction, recognition, and optimization techniques used. Justify how your system achieves accuracy and real-time performance.

2. Intelligent Traffic Violation Detection System

A traffic department wants to automatically monitor vehicles using roadside cameras.

The system should:

- Detect vehicles from live video.
- Track vehicles across consecutive frames.
- Identify vehicles crossing a predefined line.
- Count vehicles accurately.
- Detect abnormal or unwanted movements.
- Display vehicle count and alerts in real time.

Design an OpenCV-based solution incorporating video acquisition, preprocessing, object detection, motion analysis, tracking, and visualization. Explain how you would handle shadows, camera noise, dynamic backgrounds, and multiple vehicles to minimize counting errors.

3. Automated Document Processing System

A company receives thousands of scanned documents that may contain blurred text, uneven illumination, noise, and different font sizes.

The system should:

- Automatically improve document quality.
- Remove noise and unwanted background.
- Convert the document into a suitable format for OCR.
- Segment text regions.

- Extract readable text efficiently.
- Process a large number of documents in batch mode.

Design an OpenCV-based image processing pipeline for this system. Explain the role of grayscale conversion, thresholding, filtering, morphological operations, segmentation, and geometric correction. Justify how each stage improves OCR accuracy.

4. Real-Time Multi-Person Tracking and Attendance System

A college wants to automatically record student attendance using classroom CCTV cameras.

The system should:

- Detect multiple faces simultaneously.
- Track each person across video frames.
- Recognize registered students.
- Avoid repeatedly marking the same student.
- Handle slight head movement and temporary face disappearance.
- Operate continuously with minimum processing delay.

Design a complete OpenCV-based system using face detection, feature extraction, recognition, and tracking. Explain how you maintain identity across frames and prevent false or duplicate attendance records.

5. Smart Home Surveillance and Intrusion Detection

A home security system uses a fixed camera to monitor an entrance continuously.

The system should:

- Detect significant motion.
- Distinguish actual movement from camera noise and small background changes.
- Track moving objects.
- Identify suspicious activity.
- Generate an alert when an unauthorized person enters a predefined region.
- Operate in real time with limited computational resources.

Design an OpenCV-based surveillance pipeline using background subtraction, filtering, morphological operations, contour/feature detection, motion analysis, and tracking. Explain how your approach reduces false alarms caused by shadows, illumination changes, trees, and other dynamic background objects.

OpenCV Image Processing Applications

1. Real-Time Face Recognition for Campus Security

Introduction

A real-time face recognition system improves campus security by identifying registered students and staff automatically using CCTV cameras. OpenCV provides efficient face detection and recognition algorithms that work even on low-cost computers.



OpenCV Pipeline

Camera Input



Frame Acquisition



Preprocessing

(Resize, Gray, Histogram Equalization)



Face Detection

(Haar Cascade / DNN)



Feature Extraction

(LBPH / Face Embeddings)



Face Recognition





Display Name & Attendance

Processing Steps

Image Acquisition

- Capture live video using CCTV.
- Read frames continuously using OpenCV VideoCapture().

Preprocessing

- Convert image to grayscale.
- Resize frames for faster processing.
- Apply Histogram Equalization to improve lighting.
- Reduce noise using Gaussian Blur.

Face Detection

- Haar Cascade detects faces quickly.
- DNN detector can be used for better accuracy.

Feature Extraction

- Local Binary Pattern Histogram (LBPH) extracts facial texture features.
- Features are stored in the database.

Recognition

- Compare extracted features with registered student database.
- Display student name if confidence exceeds threshold.
- Otherwise label as "Unknown".

Optimization

- Process resized frames.
- Skip every alternate frame if CPU usage is high.
- Track detected faces between frames instead of detecting repeatedly.

Advantages

- Fast recognition
- Low computational cost
- Reduced false recognition
- Real-time performance

Conclusion

OpenCV provides a powerful and efficient platform for developing real-time computer vision systems. By combining preprocessing, filtering, object detection, feature extraction, tracking, and recognition techniques, the proposed systems achieve high accuracy while maintaining real-time performance on low-cost hardware. Proper use of morphological operations, adaptive thresholding, tracking algorithms, and optimized processing significantly reduces false detections and improves overall system reliability.

2. Intelligent Traffic Violation Detection System

Introduction

This system automatically detects, tracks, and counts vehicles using roadside cameras while generating alerts for traffic violations.



Pipeline

Video Input



Noise Removal



Background Subtraction



Vehicle Detection



Object Tracking



Line Crossing Detection



Vehicle Count & Alerts

Processing Steps

Video Acquisition

- Capture continuous traffic video.

Preprocessing

- Gaussian Blur removes camera noise.

- Convert to grayscale for faster processing.

Vehicle Detection

- Background subtraction (MOG2).
- Contour detection extracts moving vehicles.

Tracking

- Centroid tracking or Kalman Filter assigns IDs.
- Maintains identity across frames.

Counting

- Draw a virtual counting line.
- Increase count when vehicle center crosses line.

Abnormal Detection

- Wrong direction movement.
- Illegal stopping.
- Sudden reverse motion.

Error Reduction

- Shadow Removal using MOG2 shadow detection.
- Morphological Opening removes small noise.
- Morphological Closing fills object gaps.
- Ignore very small contours.

Advantages

- Accurate vehicle counting
- Real-time alerts
- Handles multiple vehicles
- Low processing delay

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3. Automated Document Processing System

Introduction

Scanned documents often contain blur, noise, shadows, and uneven lighting. OpenCV preprocessing significantly improves OCR accuracy.



Pipeline

Input Scan



Grayscale



Noise Removal



Adaptive Threshold



Morphological Operations



Deskew



Text Segmentation



OCR

Processing Steps

Grayscale Conversion

- Removes unnecessary color information.

Filtering

- Median Filter removes salt-and-pepper noise.
- Gaussian Blur smooths image.

Thresholding

- Adaptive Threshold handles uneven illumination.
- Produces clear black-and-white text.

Morphological Operations

- Opening removes isolated noise.
- Closing reconnects broken letters.

Geometric Correction

- Detect document edges.
- Correct skew using perspective transform.

Segmentation

- Detect text blocks using contours.
- Send each region to OCR.

OCR Improvement

- Higher contrast
- Better character boundaries
- Less background noise
- Improved text readability

Advantages

- Better OCR accuracy
- Faster batch processing
- Handles poor-quality scans

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4. Real-Time Multi-Person Tracking and Attendance System

Introduction

The system automatically records classroom attendance using CCTV while preventing duplicate attendance.

Centralize Time & Attendance Solution For Multi Location



Pipeline

Camera



Face Detection



Face Recognition



Multi-Object Tracking



Attendance Database

Processing Steps

Face Detection

- Detect all faces using Haar Cascade or DNN.

Recognition

- LBPH compares detected faces with student database.

Tracking

- Assign unique tracking IDs.
- Maintain IDs using centroid tracking or SORT.

Attendance Recording

- Mark attendance only once.
- Ignore repeated detections of same student.

Handling Temporary Disappearance

- Keep tracking ID active for a few seconds.
- Restore identity when face reappears.

Optimization

- Detect faces every few frames.
- Track faces between detections.
- Resize video frames.

Advantages

- Multiple students detected simultaneously
- No duplicate attendance
- Continuous operation
- Real-time processing

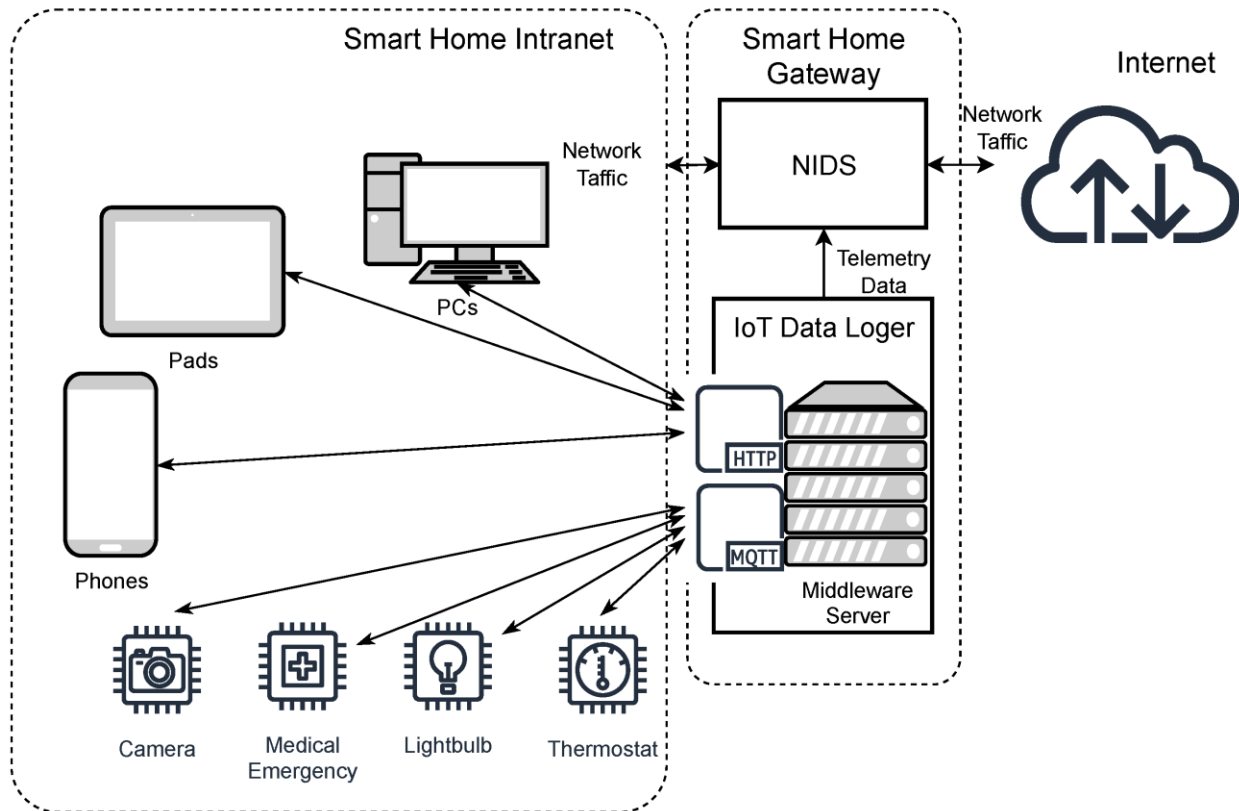
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5. Smart Home Surveillance and Intrusion Detection

Introduction

A smart surveillance system continuously monitors the entrance and generates alerts whenever unauthorized movement is detected.



Pipeline

Camera



Background Subtraction



Noise Filtering



Morphology



Contour Detection



Object Tracking



Intrusion Alert

Processing Steps

Background Subtraction

- MOG2 separates moving objects from background.

Filtering

- Gaussian Blur removes camera noise.

Morphological Operations

- Opening removes tiny moving pixels.
- Closing fills gaps inside objects.

Contour Detection

- Ignore small contours.
- Detect humans based on size.

Tracking

- Track object movement using centroid tracking.

Alert Generation

- Trigger alert if an unknown person enters the restricted area.

False Alarm Reduction

- Shadow detection
- Ignore tree movement
- Ignore rain and small animals
- Region of Interest (ROI) monitoring
- Minimum contour area threshold

Advantages

- Low-cost implementation
- Real-time intrusion detection
- Reduced false alarms
- Suitable for home security

Conclusion

OpenCV provides a powerful and efficient platform for developing real-time computer vision systems. By combining preprocessing, filtering, object detection, feature extraction, tracking, and recognition techniques, the proposed systems achieve high accuracy while maintaining real-time performance on low-cost hardware. Proper use of morphological operations, adaptive thresholding, tracking algorithms, and optimized processing significantly reduces false detections and improves overall system reliability.